

Submission PHYS1-PS2-SUB07

Question PHYS1-PS2-Q01

beginning only... show step

$$v = v_{\text{initial}} + at = 5 + 3(4) = 17 \text{ m/s}$$

[blank space]

I did not reach a conclusion

Question PHYS1-PS2-Q02

beginning only... show step

$$N = mg = 29.4 \text{ N}; \text{ friction} = \mu N = 5.9 \text{ N}$$

[blank space]

I did not reach a conclusion

Question PHYS1-PS2-Q03

beginning only... show step

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{(2gh)} = \sqrt{(2 \cdot 9.8 \cdot 6)}$$

[blank space]

I did not reach a conclusion

Question PHYS1-PS2-Q04

beginning only... show step

$$p_{\text{total, before}} = p_{\text{total, after}}$$

[blank space]

I did not reach a conclusion

Question PHYS1-PS2-Q05

beginning only... show step

$$\text{Impulse is area under } F(t): J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s}.$$

[blank space]

I did not reach a conclusion

Question PHYS1-PS2-Q06

beginning only... show step

$$v = v_{\text{initial}} + at = 4 + 2(5) = 14 \text{ m/s}$$

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I did not reach a conclusion

Question PHYS1-PS2-Q07

beginning only... show step

$$N = mg = 19.6 \text{ N}; \text{ friction} = \mu N = 3.9 \text{ N}$$

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I did not reach a conclusion

Question PHYS1-PS2-Q08

beginning only... show step

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 6}$$

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I did not reach a conclusion